Math 74: Algebraic Topology

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Problem 1. Show that for a space X , the following three conditions are equivalent:

1. Every map $S^1 \to X$ is homotopic to a constant map, with image a point.
2. Every map $S^1 \to X$ extends to a map $D^2 \to X$.
3. $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \to X$ are homotopic. [In this problem, 'homotopic' means 'homotopic without regard to basepoints'.]

Solution.

$$\begin{tikzcd}
S^1 \times [0,1] \arrow[r, "H"] \arrow[d, "\pi"'] & X \\
D^2 \arrow[ur, "f'"'] & \\
S^1 \arrow[u, "\iota"] \arrow[uur, "f"'] &
\end{tikzcd}$$

($1 \implies 2$)

Assume every map $f : S^1 \to X$ is homotopic to a constant map e_{x_0}. Let $H : S^1 \times [0,1] \to X$ be the homotopy between f and e_{x_0}, with $H(s,0) = f(s)$ and $H(s,1) = e_{x_0}(s)$. We define a quotient map,

$$\pi : S^1 \times [0,1] \to D^2, \qquad \pi(s,t) = (1-t)s$$

Note that π is continuous, surjective and maps $S^1 \times \{1\}$ to a constant point. Since H is constant on $S^1 \times \{1\}$, it induces (via π) a continuous map $f' : D^2 \to X$. Note that,

$$f' \circ \pi(s,0) = f'((1-0)s) = f'(s) = H(s,0) = f(s)$$

That is $f'(s) = f(s)$ on $s \in \partial D^2 = S^1$. Hence, every map f extends to a map f' from D^2 to X.

$(2 \implies 3)$

Assume that every map $f : S^1 \to X$ extends to a map $f' : D^2 \to X$.

Let $\iota : S^1 \to D^2$ be the natural inclusion map. Then, $f = f' \circ \iota$, and $f_\star = f'_\star \circ \iota_\star$. But,

$$\iota_\star : \pi_1(S^1, s_0) \to \pi_1(D^2, d_0)$$

is trivial because $\pi_1(D^2, d_0) = \{0\}$. Therefore $\iota_\star([\beta]) = 0$ for all $[\beta] \in \pi_1(S_1, s_0)$. Then,

$$f_\star([\beta]) = f'_\star \circ \iota_\star([\beta]) = f'_\star(0) = 0$$

Now, note that every loop $[\gamma] \in \pi_1(X, x_0)$ can be represented by some map $\gamma : S^1 \to X$, and by assumption it extendeds to some $f' : D^2 \to X$. Applying the argument above gives us $[\gamma] = 0$ and hence $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

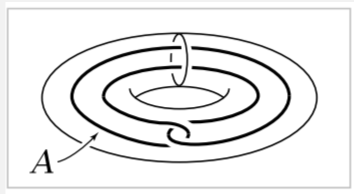
$(3 \implies 1)$

Assume $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Let $s_0 \in S^1$ be the basepoint and pick $x_0 = f(s_0)$. Since f is a map from $S^1 \to X$ it is a loop and since that every loop $[\gamma] \in \pi_1(X, x_0)$ can be represented by some map $\gamma : S^1 \to X$ we know it is homotopic to the constant loop $[e_{x_0}] \in \pi_1(X, x_0)$. And hence every map $f : S^1 \to X$ is homotopic to a constant map e_{x_0}.

Thus the three statements are equivalent.

Problem 2. Show that there are no retractions $r : X \to A$ where $X = S^1 \times D^2$ and A the circle shown in the figure.



Solution.

Note that A is homeomorphic to S^1 thus $\pi_1(A) \cong \mathbb{Z}$ with generator γ the loop that goes once around A. But the loop $\iota(\gamma)$ is homotopic to $[0] \in X$ since it can pass through itself in the solid torus. Then, the inclusion map induces the zero map on the fundamental groups. Since this is not injective a retraction $r : X \to A$ cannot exist.

Problem 3. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.

Solution.

Inductive Case: Let P be the hyperplane separating x_k from $\{x_1, \ldots, x_{k-1}\}$. Now let U be the open half-space containing x_k minus x_k, and V the open half-space containing $\{x_1, \ldots, x_{k-1}\}$ minus $\{x_1, \ldots, x_{k-1}\}$, chosen so that they overlap in a strip around the hyperplane P. That is, $U \cap V = P \times (-\epsilon, \epsilon)$ for some small $\epsilon > 0$. Then,

$$U \cup V = \mathbb{R}^n \setminus X \text{ and } U \cap V = P \times (-\epsilon, \epsilon) \cong \mathbb{R}^{n-1}$$

Then we have,

$$U \simeq \mathbb{R}^n \setminus \{x_k\} \simeq S^{n-1} \implies \pi_1(U) = \{0\}$$

(where $\pi_1(S^{n-1}) = 0$ since $n \geq 3$), and by the Inductive Hypothesis, since V is homotopy equivalent to $\mathbb{R}^n$ with $k-1$ points removed we have $\pi_1(V) = \{0\}$. Moreover, $\pi_1(P) = \{0\}$ since $P \cong \mathbb{R}^{n-1}$ and R^{n-1} is simply connected.

Then by van Kampen, we have,

$$\pi_1(\mathbb{R}^n \setminus X) \cong (\pi_1(U) \star \pi_1(V))/N = \{0\}$$

where N is trivial since $\pi_1(U \cap V) = \{0\}$. Thus the complement of a finite set of points in $\mathbb{R}^n$ is simply connected if $n \geq 3$.

Solution.

Let $X = \{x_1, \ldots, x_k\} \subset \mathbb{R}^n$. We prove $\pi_1(\mathbb{R}^n \setminus X) = \{0\}$ by induction on k.

Base Case: $k = 1$.

We know that $\mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ by deformation retraction, and that $\pi_1(S^{n-1}) = \{0\}$ since $n - 1 \geq 2$. Hence, $\mathbb{R}^n \setminus \{x\}$ is simply connected.

Inductive Hypothesis. Assume that for any set of $k-1$ points, $\pi_1(\mathbb{R}^n \setminus X) = \{0\}$.

Inductive Step. Since the x_i are distinct, choose a hyperplane $P \cong \mathbb{R}^{n-1}$ separating x_k from $\{x_1, \ldots, x_{k-1}\}$. Define

$$U = \big(H^+ \cup (P \times (-\varepsilon, \varepsilon))\big) \setminus \{x_k\}, \qquad V = \big(H^- \cup (P \times (-\varepsilon, \varepsilon))\big) \setminus \{x_1, \ldots, x_{k-1}\},$$

where H^+ is the open half-space on the side of x_k, H^- is the open half-space on the side of $\{x_1, \ldots, x_{k-1}\}$, and $\varepsilon > 0$ is picked to be small enough that the strip $P \times (-\varepsilon, \varepsilon)$ contains none of the x_i. Then U and V are open sets satisfying

$$U \cup V = \mathbb{R}^n \setminus X, \qquad U \cap V = P \times (-\varepsilon, \varepsilon) \cong \mathbb{R}^{n-1}.$$

Note that since $n \geq 2$, removing finitely many points from an open subset of $\mathbb{R}^n$ preserves path-connectivity, so U, V, and $U \cap V \cong \mathbb{R}^{n-1}$ are all path-connected. Fix a basepoint $x_0 \in U \cap V$. We now compute the fundamental groups of each piece:

$$U \simeq \mathbb{R}^n \setminus \{x_k\} \simeq S^{n-1} \implies \pi_1(U) = \{0\},$$

where $\pi_1(S^{n-1}) = \{0\}$ since $n \geq 3$. By the inductive hypothesis, $V \simeq \mathbb{R}^n \setminus \{x_1, \ldots, x_{k-1}\}$ gives $\pi_1(V) = \{0\}$. Moreover, $U \cap V \cong \mathbb{R}^{n-1}$ is simply connected, so $\pi_1(U \cap V) = \{0\}$.

Then by the Seifert–van Kampen theorem applied to the cover $\{U, V\}$,

$$\pi_1(\mathbb{R}^n \setminus X) \cong \big(\pi_1(U) \star \pi_1(V)\big)/N,$$

Since $\pi_1(U \cap V) = \{0\}$, the subgroup N is trivial, and so

$$\pi_1(\mathbb{R}^n \setminus X) \cong \{0\} \star \{0\} = \{0\}.$$

Thus the complement of any finite set of points in $\mathbb{R}^n$ is simply connected for $n \geq 3$.

Problem 4. Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 - X)$.

Solution.

We can deformation retract $\mathbb{R}^3$ onto S^2 by the family of maps $\phi_t(x) = x/|x|^t$ where $t \in [0,1]$ and x is a point in $\mathbb{R}^3$. Now note that each line through the origin in $\mathbb{R}^3$ intersects S^2 in exactly two antipodal points. Since $\phi_t(x)$ is a multiple of x, if $x \notin X$ then $\phi_t(x) \notin X$, so, $\mathbb{R}^3 \setminus X$ deformation retracts to $S^2 \setminus \{p_1, \ldots, p_{2n}\}$ where $p_1, \ldots, p_{2n}$ are the antipodal points the lines in X would've mapped onto. Hence,

$$\mathbb{R}^3 \setminus X \simeq S^2 \setminus \{p_1, \ldots, p_{2n}\}$$

Now, recall that removing a point from S^2 via stereographic projection gives us $\mathbb{R}^2$. We. will show that $\pi_1(\mathbb{R}^2 \setminus \{q_1, \ldots, q_{2n-1}\}) \cong \star\mathbb{Z}^{2n-1}$ via induction on the number of points. For convenience, let $2n - 1 = m$.

Base Case: $m = 1$.

$\mathbb{R}^2 \setminus \{q\} \simeq S^1$, so we have $\pi_1(\mathbb{R}^2 \setminus \{q\}) \cong \mathbb{Z} = \star\mathbb{Z}^1$.

Inductive Hypothesis: Assume that $\pi_1(\mathbb{R}^2 \setminus \{q_1, \ldots, q_t\}) \cong \star\mathbb{Z}^t$, for $t < m$.

Inductive Step: Since the q_i points are distinct, choose a line L separating q_m from $\{q_1, \ldots, q_{m-1}\}$. Define

$$U = \big(H^+ \cup (L \times (-\varepsilon, \varepsilon))\big) \setminus \{q_m\}, \qquad V = \big(H^- \cup (L \times (-\varepsilon, \varepsilon))\big) \setminus \{q_1, \ldots, q_{m-1}\},$$

where H^+ is the open half-space on the side of q_m, H^- is the open half-space on the side of $\{q_1, \ldots, q_{m-1}\}$, and $\varepsilon > 0$ is picked to be small enough that the strip $L \times (-\varepsilon, \varepsilon)$ contains none of the q_i. Then U and V are open sets satisfying

$$U \cup V = \mathbb{R}^2 \setminus \{q_1, \ldots, q_m\}, \qquad U \cap V = L \times (-\varepsilon, \varepsilon) \cong \mathbb{R}^1.$$

Note that removing finitely many points from an open subset of $\mathbb{R}^2$ preserves path-connectivity, and L is a line and is simply connected. So U, V, and $U \cap V \cong \mathbb{R}^1$ are all path-connected. Fix a basepoint $x_0 \in U \cap V$. We now compute the fundamental groups of each piece:

$$U \simeq \mathbb{R}^2 \setminus \{q_m\} \simeq S^1 \implies \pi_1(U) = \mathbb{Z},$$

Then, by the inductive hypothesis, $V \simeq \mathbb{R}^2 \setminus \{q_1, \ldots, q_{m-1}\}$ gives $\pi_1(V) = \star\mathbb{Z}^{m-1}$, and we know that $U \cap V \cong \mathbb{R}^1$ is simply connected, so $\pi_1(U \cap V) = \{0\}$.

Then by the Seifert–van Kampen theorem applied to the cover $\{U, V\}$,

$$\pi_1(\mathbb{R}^2 \setminus \{q_1, \ldots, q_m\}) \cong \big(\pi_1(U) \star \pi_1(V)\big)/N,$$

Since $\pi_1(U \cap V) = \{0\}$, the subgroup N is trivial, and so

$$\pi_1(\mathbb{R}^2 \setminus \{q_1, \ldots, q_m\}) \cong \mathbb{Z} \star \mathbb{Z}^{m-1} = \star\mathbb{Z}^m.$$

So we have,

$$\mathbb{R}^3 \setminus X \simeq S^2 \setminus \{p_1, \ldots, p_{2n}\} \simeq \mathbb{R}^2 \setminus \{q_1, \ldots, q_{2n-1}\}$$

And therefore,

$$\pi_1(\mathbb{R}^3 \setminus X) \cong \pi_1(S^2 \setminus \{p_1, \ldots, p_{2n}\}) \cong \pi_1(\mathbb{R}^2 \setminus \{q_1, \ldots, q_{2n-1}\}) = \star\mathbb{Z}^{2n-1}.$$